

CN Unit 1 — Introduction & Physical Layer

Complete Notes

Computer Networks (CN) · Unit 1 · Introduction & Physical Layer · 2025

Overview

These notes cover **Introduction & Physical Layer**. From bits on the wire to HTTP — layered network architectures, routing, transport protocols and the applications on top. Each section states the core idea, a worked intuition and the points examiners look for.

1. OSI Reference Model

The OSI model has seven layers: physical, data link, network, transport, session, presentation and application.

Why it matters: OSI Reference Model appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of osi reference model in one line.
- A small example showing osi reference model in practice.
- Common mistakes students make and how to avoid them.

2. TCP/IP Model

The TCP/IP stack collapses OSI into link, internet, transport and application layers.

Why it matters: TCP/IP Model appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of tcp/ip model in one line.
- A small example showing tcp/ip model in practice.
- Common mistakes students make and how to avoid them.

3. Network Topologies

Bus, star, ring, mesh and hybrid topologies differ in cost, fault tolerance and performance.

Why it matters: Network Topologies appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of network topologies in one line.
- A small example showing network topologies in practice.
- Common mistakes students make and how to avoid them.

4. Transmission Media

Twisted pair, coaxial cable, optical fibre and wireless media carry signals with different bandwidths.

Why it matters: Transmission Media appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of transmission media in one line.
- A small example showing transmission media in practice.
- Common mistakes students make and how to avoid them.

5. Multiplexing

FDM, TDM and WDM share a single channel among multiple signals.

Why it matters: Multiplexing appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of multiplexing in one line.
- A small example showing multiplexing in practice.
- Common mistakes students make and how to avoid them.

6. Switching Techniques

Circuit switching reserves a path; packet switching forwards independent packets.

Why it matters: Switching Techniques appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of switching techniques in one line.
- A small example showing switching techniques in practice.
- Common mistakes students make and how to avoid them.

Quick revision

OSI Reference Model · TCP/IP Model · Network Topologies · Transmission Media · Multiplexing · Switching Techniques